

Akhil Prasad Chinthala

AI / ML Engineer | LLM & Computer Vision Systems

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SUMMARY

Final-year IT student (CGPA 8.64) building LLM features as an AI Engineering Intern at AMIK Technologies and writing agentic coding benchmarks for frontier-model evaluation as a freelance AI trainer at Handshake AI. Works across Python backends and applied AI: FastAPI model serving, retrieval-augmented generation, LLM evaluation and real-time computer vision (CNN emotion recognition at 70.6% on FER-2013, about 8 ms per frame on CPU). Looking for AI/ML or Python backend roles on production LLM and vision systems.

EXPERIENCE

AI Engineering Intern

AMIK Technologies

Apr 2026 - Present

Hyderabad, India

- Build production retrieval-augmented generation (RAG) pipelines over internal documents (chunking, embedding generation and vector similarity search), replacing manual lookup with grounded answers that cite their sources.
- Design and version prompt templates with structured outputs and function calling, so LLM responses come back as schema-validated JSON that downstream services consume directly.
- Ship an automated evaluation harness that scores retrieval relevance, answer faithfulness and latency across model and prompt versions, so releases are regression-tested instead of reviewed by eye.
- Serve models behind FastAPI endpoints with token streaming, per-request cost logging and provider fallback for when rate limits hit.
- Cut inference cost and latency with chunk-size tuning, semantic caching and context compression, and work with the backend team to ship features into the live product.

Freelance AI Trainer, Agentic Benchmark Authoring

Handshake AI

Jul 2026 - Present

Remote

- Write terminal-based benchmark tasks that evaluate frontier AI coding agents: 102 tasks across 10 domains, including systems infrastructure, model training and ML pipelines, data processing/ETL, databases and interactive simulation.
- Each task ships as a reproducible package (a natural-language spec, a Dockerised environment, an automated test-based verifier and a working reference solution) and has to pass automated QA gates and human review.

Software Development Intern (Python)

Oasis Infobyte

Oct 2025 - Nov 2025

Remote

- Built five Python applications on a structured internship track, each specified, built and demoed on its own.
- They included a speech-recognition voice assistant that turns spoken intent into system commands, a REST-integrated weather CLI and a multi-client TCP chat server that handles concurrent sessions.

PROJECTS

RBI FEMA RAG: Evaluated Regulatory Q&A | Python, sentence-transformers, Ollama

2026

- Built a fully local RAG system over RBI's foreign exchange Master Directions (~570 pages) with hybrid BM25 + dense retrieval, cross-encoder reranking and cited answers, evaluated against 186 question-answer pairs from RBI's own published FAQs.
- Replaced an unreliable LLM relevance judge (precision 0.12 to 0.15) with a calibrated embedding judge (95% agreement on checked labels) to build pooled relevance labels, with bootstrap confidence intervals and paired significance tests throughout. Best retrieval: **83% hit@5 vs 59% for BM25**.
- Found that the LLM answered 67% of questions the documents could not answer; a cross-validated retrieval-confidence gate raised **correct refusals from 33% to 75%**.

CAG Real-Time Emotion Tracker | PyTorch, OpenCV, scikit-learn, Streamlit, Python

2024 - 2026

- Trained a 5M-parameter CNN with squeeze-and-excitation attention on FER-2013 to **70.6% test accuracy across 7 emotion classes** (human agreement $\approx$ 65%), up from 38% for the team's hand-crafted-feature baseline.
- Replaced the classifier head with a cache-augmented lookup (one matrix multiply against in-memory per-class prototype embeddings), keeping the full webcam pipeline at **about 8 ms per frame on CPU**.
- Led a 3-person team from architecture through delivery, shipped a Streamlit dashboard and presented the results to a faculty review panel.

- E2EE Chat: End-to-End Encrypted Group Chat** | *Python, cryptography, PyQt5, Sockets* 2026
- Rebuilt the internship chat app with end-to-end encryption: X25519/Ed25519 identities generated on each device, AES-256-GCM messages with per-recipient key wrapping under ephemeral X25519 keys, and Ed25519-signed envelopes. The relay server only stores ciphertext it cannot read.
 - Key pinning and Signal-style safety numbers guard against a malicious server; **14 tests** show that tampering, forged senders and a **server key-swap attack** are rejected. Also fixed an HTML-injection bug in the original GUI.
- Full-Stack Booking Platform with Voice AI Assistant** | *React, Vite, Node.js, Express, Web Speech API* 2025
- Built an end-to-end web app (React/Vite frontend on an Express REST API) that runs in production for a real business client, a dental clinic, and handles real customer bookings.
 - Integrated a voice-enabled AI assistant (*DentalBot*) that answers the client's service FAQs and turns visitors into bookings.
 - Automated the booking workflow: appointments go through the REST API, sync to Google Sheets for the client's staff and trigger email confirmations.
- Hand-Gesture Media Controller** | *MediaPipe, OpenCV, Python, AppleScript* 2024
- Mapped MediaPipe hand-landmark detection to macOS volume and playback controls, so media can be controlled without touching anything.
 - Sustained **30 FPS with sub-20 ms gesture-response latency** by optimising the landmark-to-action inference loop.
- Developer Portfolio** (akhil-prasad09.github.io) | *Next.js, React, TypeScript* 2026
- Single-page developer portfolio with animated sections for projects, benchmarks and experience, exported as a static Next.js site and deployed to GitHub Pages.
- Green Basket: E-Commerce Web App** | *React, JavaScript, CSS3* 2024
- Responsive grocery storefront built in React, with dynamic product filtering, cart state management and a full checkout flow.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Java, C, C++

LLM & Generative AI: Retrieval-Augmented Generation (RAG), prompt engineering, embeddings & vector search, function calling / structured outputs, LLM evaluation, semantic caching

Machine Learning & CV: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, MediaPipe, NumPy, Pandas, fine-tuning

Deployment & Tooling: FastAPI, REST APIs, Git & GitHub, Linux (Ubuntu), model serving, latency & cost profiling

Web & Data: React, Vite, Node.js, Express, MySQL, MongoDB, vector databases

Foundations: Deep Learning, Computer Vision, NLP, Data Structures & Algorithms, Operating Systems, Networks

Spoken languages: Telugu (Native), English (Fluent), Hindi (Conversational)

EDUCATION

Matrusri Engineering College , B.Tech in Information Technology, CGPA 8.64	2023 - 2027
Sri Chaitanya Junior Kalasala , Higher Secondary in Science & Mathematics, 93%	2023
The Hyderabad Public School , Secondary School, 89%	2021